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Black Holes Discovered and Rejected

in which Einstein's laws
of warped spacetime
predict black holes,
and Einstein rejects the prediction

"The essential result of this investigation," Albert Einstein wrote in a technical paper in 1939, "is a clear understanding as to why the 'Schwarzschild singularities' do not exist in physical reality." With these words, Einstein made clear and unequivocal his rejection of his own intellectual legacy: the black holes that his general relativistic laws of gravity seemed to be predicting.

Only a few features of black holes had as yet been deduced from Einstein's laws, and the name "black holes" had not yet been coined; they were being called "Schwarzschild singularities." However, it was clear that anything that falls into a black hole can never get back out and cannot send light or anything else out, and this was enough to convince Einstein and most other physicists of his day that black holes are outrageously bizarre objects which surely should not exist in the real Universe. Somehow, the laws of physics must protect the Universe from such beasts.

What was known about black holes, when Einstein so strongly rejected them? How firm was general relativity's prediction that they do exist? How could Einstein reject that prediction and still maintain confidence in his general relativistic laws? The answers to these questions have their roots in the eighteenth century.